# VIT BHOPAL UNIVERSITY

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# FUNDAMENTALS IN AI AND ML

## PROJECT REPORT

SLOT: A14+D11+D12

## " CyberGuard AI "

vityarthi
YOUR LEARNING DESTINATION


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# Introduction

In today's digital world, communication through emails and messages has become very common. However, along with convenience, there has been a significant rise in spam messages, phishing emails, and online scams. These messages often try to trick users into sharing personal or financial information.

Many users are not able to easily identify whether a message is safe or harmful. This creates a need for a system that can automatically analyze and classify messages.

CyberGuard AI is developed to address this issue. It is a simple and effective system that uses machine learning and basic rule-based checks to detect whether a message or email is safe, suspicious, or spam.

# Problem Statement

Spam messages and fraudulent emails are increasing rapidly and often go unnoticed by users. Many people are unable to differentiate between genuine and malicious messages, leading to financial loss and data theft.

There is a need for a system that can:

- Automatically detect spam messages
- Analyze complete emails, not just short text
- Provide clear and understandable results to users

# Objectives

The main objectives of this project are:

- To develop a spam detection system using Machine Learning
- To analyze both messages and full email content
- To extract meaningful information from raw text
- To classify messages into Safe, Suspicious, or Spam
- To improve user awareness about cyber threats

# Methodology

The project follows a structured approach:

1.Data Collection

A spam dataset (spam.csv) is used containing labeled messages (ham and spam).

2.Data Preprocessing

- Convert text to lowercase
- Remove special characters
- Handle links and numbers
- Clean unwanted symbols

3.Feature Extraction

TF-IDF (Term Frequency-Inverse Document Frequency) is used to convert text into numerical form.

4.Model Training

A Multinomial Naive Bayes algorithm is used to train the model.

5.Prediction

The model predicts whether the message is spam or not.

6.Rule-Based Analysis

Additional checks are applied:

- Suspicious keywords
- Number of links
- Capital words

7.Final Decision

A combined result is generated as:

- Safe
- Suspicious
- Spam

# System Design

The system follows this flow:

Input → Processing → Output

- Input: User enters message or email
    - Processing:Extract email body
    - Clean text
    - Apply ML model
    - Apply rule-based checks
- Output: Final classification with analysis report and graph

# Implementation

The system is implemented using Python.

Key components include:

- clean_text()
- Cleans and normalizes input text, removes unwanted characters
- extract_body()
- Extracts only the meaningful part of an email, ignoring headers and signatures
- rule_based_check()
- Detects suspicious keywords like "free", "win", "urgent", etc.
    - Machine Learning ModelTF-IDF Vectorizer
    - Multinomial Naive Bayes classifier
    - VisualizationMatplotlib is used to generate summary graphs

# Results

The system successfully classifies messages into:

- Safe
- Suspicious
- Spam

It provides:

- Spam probability
- Detected suspicious words
- Number of links and numeric patterns
- Final verdict

The model achieves good accuracy on the dataset and performs well on real-world email inputs.

# Challenges Faced

While working on this project, a few challenges were faced:

- Handling full email inputs instead of just single-line messages
- Managing special characters that caused errors
- Avoiding multiple graphs for one message
- Extracting only meaningful content from emails
- Balancing machine learning results with rule-based logic

These challenges were solved through testing and improving the code.

# Applications

This system can be used in many real-life situations:

- Email filtering systems
- Spam detection in messaging apps
- Cybersecurity awareness tools
- Educational purposes

# Limitations

- Accuracy depends on dataset quality
- Cannot detect highly advanced or new types of spam
- Does not analyze attachments or images
- Limited to text-based messages

# Future Enhancements

The project can be improved further by:

- Creating a web-based interface
- Connecting it with real email services
- Using advanced machine learning or deep learning models
- Adding support for multiple languages

# Conclusion

- CyberGuard AI is a simple yet effective spam detection system. It shows how machine learning and basic logic can be combined to solve real-world problems.
- The project helped in understanding important concepts like text processing, machine learning, and cybersecurity. It can be further improved and expanded into a more advanced system in the future.

# References


- Kaggle Spam Dataset
- Scikit-learn Documentation
- Python Official Documentation
- Matplotlib Documentation